

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Advanced Machine Learning, Advanced NLP, Optimization in CS

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, JavaScript, TypeScript, C++, Go, Java, SQL, Bash

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services, authentication, rate limiting

Testing & Observability

pytest, integration & regression testing, automation scripting, pipeline instrumentation, production log monitoring, Spark UI profiling

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Databases & Web

MongoDB, PostgreSQL, Redis, SQLite, React, Electron

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Owned backend reliability and observability across the eTriage stack: monitored production logs during releases and client integrations to triage and ship fixes for tenant-specific incidents, hardened authentication and rate limiting, added integration tests for regression protection, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings**.
- Architected internal full-stack tooling (**React, Node.js, MongoDB**) used daily by the AI team, replacing manual spreadsheet workflows with secure role-based REST APIs and streamlining a repetitive operational process; cut manual work by **100+ hours/month** and improved throughput by **60%**.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety, maintainability, and overall code quality.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Hardened a Python/FastAPI web-discovery service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation to keep behavior predictable and observable.
- Designed the pipeline as discrete, testable stages, typed retrieval planning, page rerank, deterministic extraction with LLM fallback, and output verification, instead of a single opaque call, returning cell-level provenance for every result.

Spark ETL Performance Optimization - NYC TLC Trip Data

github.com/Deva-1903/Spark-ETL-Optimization

Python, PySpark, Spark UI, SQL

- Cut end-to-end ETL runtime **~25%** (17 to 19 min down to 14.3 min) on a **~1.4B-row** dataset by replacing shuffle joins with broadcast joins (minutes to ~0.05s), enabling adaptive query execution and skew-join handling, and tuning shuffle partitions, schema-aware reads, and partitioning.
- Used **Spark UI** stage-level metrics and timings to localize shuffle and skew bottlenecks and document a repeatable optimization playbook, cutting the analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x**.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform desktop application** integrating camera, microphone, UI, and backend services into an on-device pipeline (MediaPipe hand tracking, gesture classification, whisper.cpp Automatic Speech Recognition (ASR), and local Text-To-Speech (TTS)), a user-friendly tool prototyped, designed, and shipped under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** to run fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling local video and audio streams on device.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.